

# Waves

WAVES ARE VIBRATIONS THAT TRANSFER ENERGY.

Many different types of energy travel in waves. Sound waves carry noises through air, while seismic waves travel inside the Earth and cause earthquakes.

## What is a wave?

Waves are vibrations that transfer energy as they travel. Some energy waves—sound waves, for example—need to travel through a medium, such as water or air. The medium does not travel with the wave, but moves back and forth as energy is passed through it, similar to the way a “wave” travels around a sports stadium as people move up and down in their seats. There are two main types of wave: transverse and longitudinal.

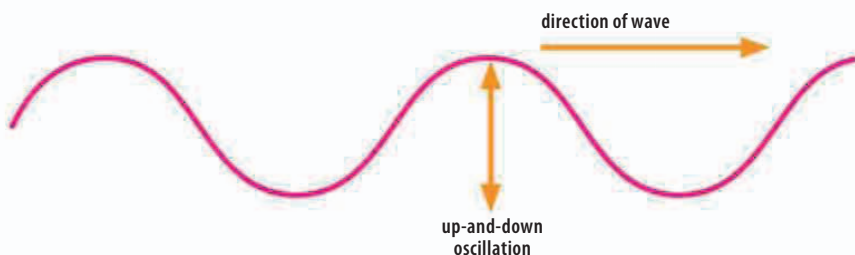
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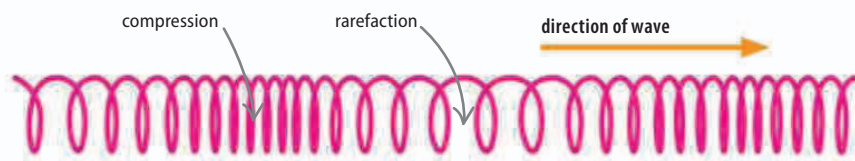
#### Seismic waves

Seismic waves are caused by the movement of rocks underground. As the vibrations travel up to the Earth's surface, they can produce earthquakes. The huge amounts of energy in seismic waves can be detected many thousands of kilometers away by sensitive instruments called seismometers.



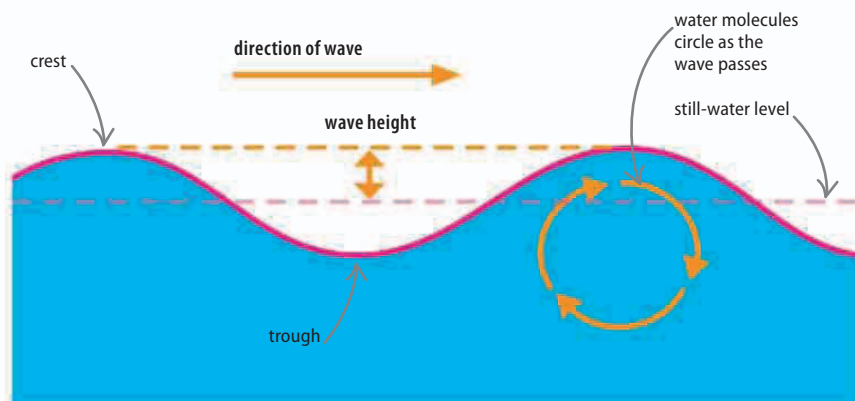
#### ◀ Transverse wave

Light and other electromagnetic waves are transverse waves. This type of wave oscillates (vibrates) up and down at right angles (transversely) to the direction of travel of the wave, following an S-shaped path.



#### ◀ Longitudinal wave

Sound energy travels in longitudinal waves. The effect is like releasing a stretched spring and watching the energy travel along the coils, squeezing them together (sections called compressions) and stretching them apart (sections called rarefactions). Sound moves through air by pushing and pulling air molecules in a similar pattern of compressions and rarefactions.



#### ◀ Ocean wave

Ocean waves are formed by the action of the wind pushing against the surface of the water, and have qualities of both of the wave types above. At the surface, water rises and falls between its highest point (crest) and lowest point (trough), equidistant to the still-water level. As the wave passes, the water molecules below the surface do not move forward but loop in circles.